

Boyao Liao — Curriculum Vitae

bxl307@student.bham.ac.uk & BoyaoLiao@stu2023.jnu.edu.cn (+86) 19807907696

RESEARCH INTEREST

Machine learning, Truthful Neural Network,
Optimization Theory

EDUCATION

Jinan University – University of Birmingham Joint Institute

B.Sc. in Mathematics with Applied Mathematics

Guangzhou, CHN

September 2023- now

- GPA: 4.14/4.25(Criterion of University of Birmingham, above 75 is A+)
- Rank: 3/68
- Centesimal grade average: 87.4(for math and programming courses)(Criterion of University of Birmingham, above 75 is A+)
- GPA: 4.29/5(Criterion of Jinan University, 90-100 is A)
- Rank: 3/68
- Centesimal grade average: 92.9(Criterion of Jinan University, 90-100 is A)

AWARDS & HONORS

- **National Scholarship**, Ministry of Education of the People's Republic of China, 2024
- **Outstanding Student**, Jinan University, 2024

RESEARCH EXPERIENCE

A research on loss landscape of Deep Learning

2025-now

- It focuses on landscape feature during the training process of Deep Learning.
- Landscape shows us some features especially of Hessian matrix during the training. It provides hints for optimization, learning rate sensitivity, local minima and generalization, and training dynamics:

A new Dynamic and Interpretable Neural Network

2024

- It focuses on deep learning with data preprocessing, dynamic network tuning, and PDE regularization.
- It extracts the important features of the picture through preprocessing operations on the image data and utilizes a single-layer dynamic network to achieve the classification of the picture.
- It simplifies the traditional network structure, improves interpretability and robustness, and also performs well compared to other models.

Intelligent pH value detection device based on image recognition technology

2024

- It focuses on the invention of a new device for distinguishing pH by image recognition of the color of pH test paper.
- The present invention provides a pH intelligent detection device that can not only be capable of realizing high-precision pH measurement, but also capable of real-time transmission of test results through a wireless data transmission module
- It is a compact, low-cost, and easy-to-operate pH intelligent detection device, the invention ensures the accuracy of the detection and improves the detection speed and user experience.

SKILLS

Programming	Matlab, LaTeX, Python, Pytorch, R, C, Java(a little)
Languages	Chinese, English
Software	Office

PUBLICATIONS

- **A Study of Dynamic Neural Network Framework Based on Data Preprocessing and PDE Integration(accepted)**
Boyao Liao* and Zujian, Wu and Lie, Luo and Jenny, Jiang and Gang, Lu
The 10th International Conference on Computer and Communication Systems Chengdu, China — April 18-21, 2025
, 2025. **IEEE & EI & Scopus Conference**

Patents

- **A kind of intelligent pH value detection device based on image recognition technology**
Boyao Liao and Zujian, Gang, Lu et.al

SELECTED COURSES

- **Postgraduate(Self-study):**
High-dimensional Probability, Convex optimization
- **Undergraduate(ALL mathematics and programming courses are above 90):**
Vector,Geometry& Linear Algebra (99), Probability& Statistics (98) Linear Algebra& Linear Programming(97), Sequences and Series (98),Multiple variable Calculus(95), Real Analysis(94), Analytical Geometry(94), Algebra&Combinatorics(94), Introduction to Data Analysis (94),C Programming(90)

REFERENCES

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| ○ Dr. Zujian Wu
Associate professor, Jinan University – University
of Birmingham Joint Institute
Jinan University,Guangzhou, CHN
Email: Zujian.Wu@jnu.edu.cn | ○ Dr. Gang Lu
Associate professor, Jinan University-College of En-
vironment and Climate
Jinan University, Guangzhou, CHN
Email: tlugang@jnu.edu.cn |
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